

Trigonometry
Quadrant system

Quadrant system :

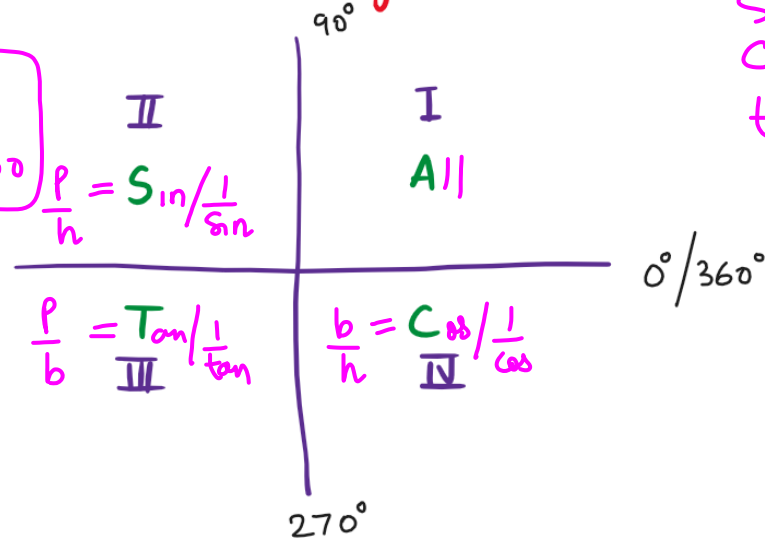
$\sin\theta \rightarrow$ (-ve) $-\frac{p}{h}$

$\cos\theta$ +ve (0-90, 270-360)

$\frac{p}{b} = \tan\theta = +ve = \frac{+ve}{+ve} = \frac{-ve}{-ve}$

h = +ve

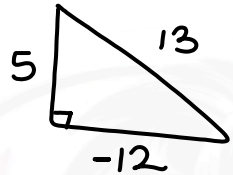
S → Csc
C → Sec
t → Cot



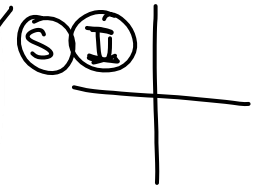
If $\sin\theta = 5/13$ and θ is in the second quadrant, find the value of $\sin\theta + \cos\theta$.

यदि $\sin\theta = 5/13$ है और θ दूसरा चतुर्थांश में है, तो $\sin\theta + \cos\theta$ का मान ज्ञात कीजिए।

a) -7/13
b) 7/3
c) 14/13
d) -14/13



$\frac{5}{13} - \frac{12}{13} = -\frac{7}{13}$



2 If $sinA = \frac{4}{5}$ and A lies in ~~2nd quadrant~~, find the value of $(1 + tanA)(1 - tanA)$.

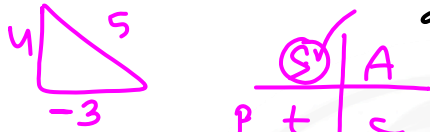
यदि $sinA = \frac{4}{5}$ और A दूसरे चतुर्थांश में स्थित है, तो $(1 + tanA)(1 - tanA)$ का मान ज्ञात कीजिए।

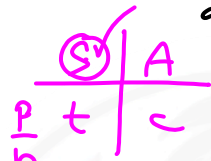
a) $-\frac{9}{25}$

b) $\frac{16}{25}$

c) $\frac{1}{9}$

d) $-\frac{7}{9}$





$\left(1+\left(\frac{-4}{3}\right)\right)\left(1-\left(\frac{-4}{3}\right)\right)$

$1 - \frac{16}{9} = -\frac{7}{9}$

$1 - \tan^2 A$

$\cot^2 A - \csc^2 A + \sec^4 A$

$(-5)^2 = 5^2$

3 If $sinA = \frac{5}{13}$ and A lies in 2nd quadrant, then find the value of $sinA \times secA$.

यदि $sinA = \frac{5}{13}$ और A दूसरे चतुर्थांश में स्थित है, तो $sinA \times secA$ का मान ज्ञात कीजिए।

a) $\frac{5}{12}$

b) $-\frac{5}{12}$

c) $\frac{25}{12}$

d) $\frac{25}{13}$

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4 If $\sin\theta = \frac{7}{25}$ and θ lies in the second quadrant, find the value of tan θ.

यदि $\sin\theta = \frac{7}{25}$ हो और θ द्वितीय चतुर्थांश में स्थित हो, तो tan θ का मान ज्ञात कीजिए।

a. $-\frac{7}{24}$

b. $-\frac{8}{25}$

c. $-\frac{5}{24}$

d. $-\frac{7}{25}$

$-\frac{7}{24}$

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